SP-C36: Cafe Orders — R Time Series Analysis

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Activity 8

1. Aim

To analyze daily cafe orders using time-series plots, seasonal decomposition, seasonal consistency, autocorrelation, and trend analysis in R.

2. Task 1 — Plot the series and identify the trend

Code:

```
set.seed(7908)

m <- 7

n <- m * 11

t <- 1:n

trend <- 206.1 + 3.117 * t

orders <- round(trend * (1 + 0.28 * cos(2 * pi * (t - 3) / m)) +

                rnorm(n, 0, 8.8), 0)

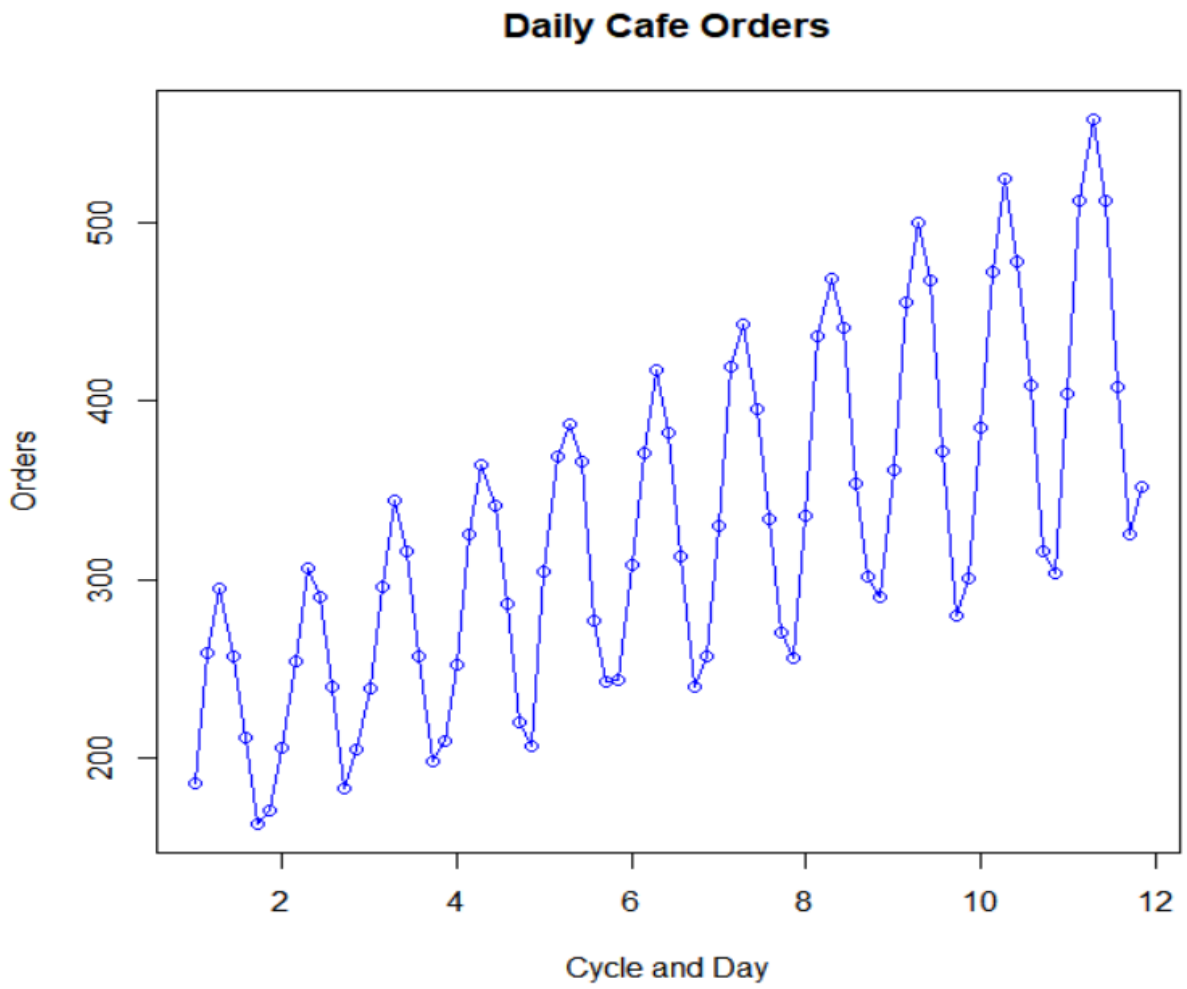
x <- ts(orders, frequency = m, start = c(1, 1))
```

```
plot(x, type = "o", col = "blue",
     main = "Daily Cafe Orders",
     xlab = "Cycle and Day",
     ylab = "Orders")

fit <- lm(orders ~ t)
slope <- coef(fit)[2]

cat("Number of observations:", n, "\n")
cat("Cycle length:", m, "\n")
cat("Number of full cycles:", n / m, "\n")
cat("Trend slope per day:", slope, "\n")
```

Output



Answer:

- Number of observations: $n = 77$
- Cycle length: 7 observations.
- Number of full cycles: $77/7 = 11$.
- Overall trend: Rising, because the underlying trend increases by 3.117 orders per day. Report the exact fitted slope printed by R.

Interpretation: The plot shows daily cafe orders with a repeating weekly pattern and an overall upward trend.

3. Task 2 — Seasonal pattern

Code:

```
dc <- decompose(x, type = "multiplicative")
seasonal <- dc$figure
names(seasonal) <- paste("Position", 1:7)

print(seasonal)

peak_day <- which.max(seasonal)
trough_day <- which.min(seasonal)
seasonal_swing <- max(seasonal) - min(seasonal)

cat("Peak position:", peak_day, "\n")
cat("Peak seasonal value:", seasonal[peak_day], "\n")
cat("Trough position:", trough_day, "\n")
cat("Trough seasonal value:", seasonal[trough_day], "\n")
cat("Seasonal swing:", seasonal_swing, "\n")

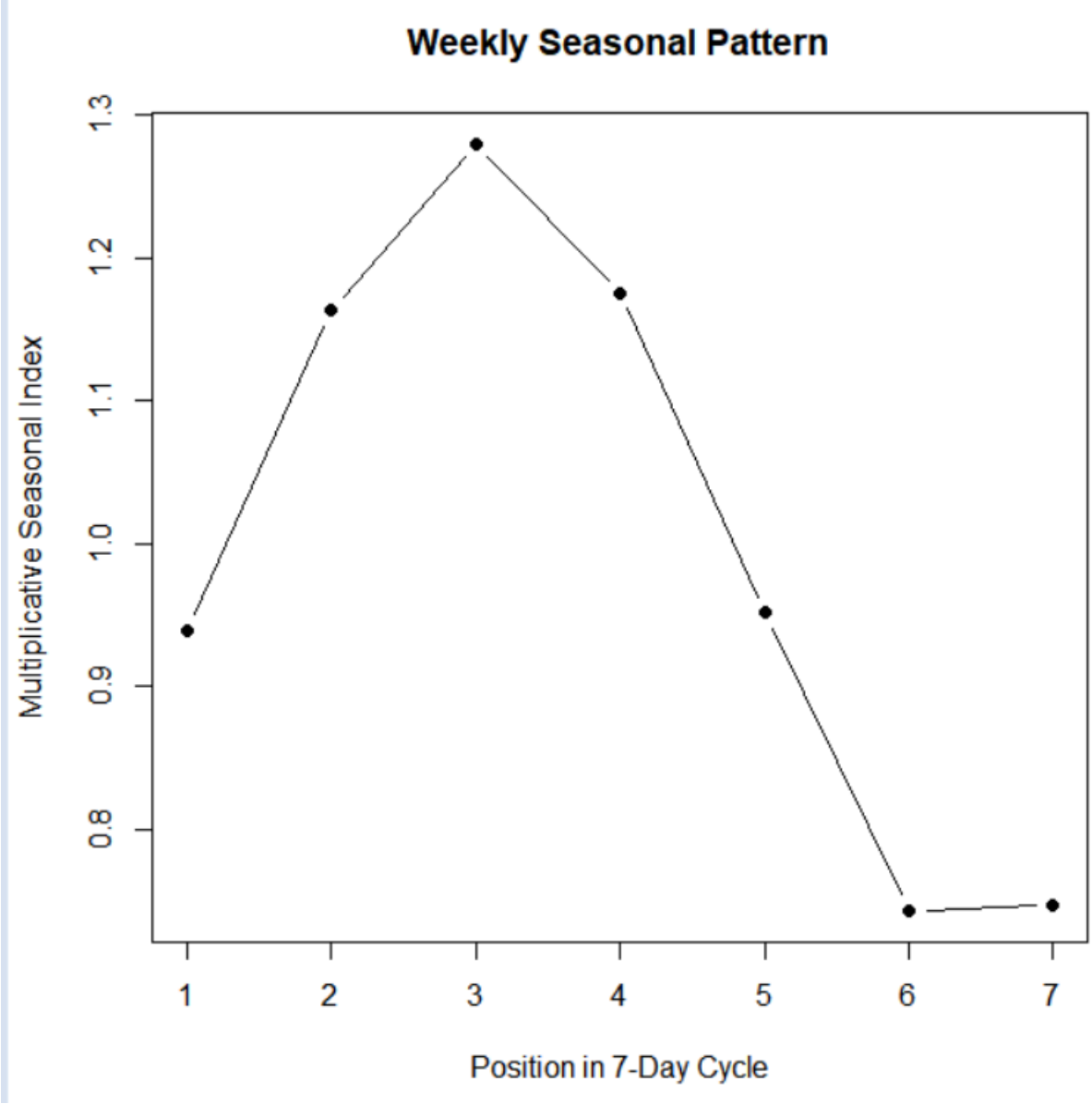
plot(seasonal, type = "b", pch = 19,
```

main = "Weekly Seasonal Pattern",

xlab = "Position in 7-Day Cycle",

ylab = "Multiplicative Seasonal Index")

Output



Answer and interpretation:

- Peak day: the position with the highest seasonal index.
- Trough day: the position with the lowest seasonal index.
- Seasonal swing: highest seasonal index minus lowest seasonal index.

The exact positions and values are printed by the code. Position 1 to 7 means the observation's position within the cycle; actual weekdays were not specified in the data.

4. Task 3 — Check seasonal consistency

Code:

```
cycle <- rep(1:11, each = 7)

position <- rep(1:7, times = 11)


season_data <- data.frame(

  orders = orders,

  cycle = factor(cycle),

  position = position

)


library(ggplot2)


ggplot(season_data,
```

```
      aes(x = position, y = orders,
          group = cycle, colour = cycle)) +
  geom_line() +
  geom_point() +
  labs(title = "Seasonal Plot of Cafe Orders",
       x = "Position in 7-Day Cycle",
       y = "Orders",
       colour = "Cycle") +
  theme_minimal()

peak_day <- which.max(seasonal)

peak_positions <- sapply(1:11, function(i) {
  start <- (i - 1) * 7 + 1
  cycle_orders <- orders[start:(start + 6)]
  which.max(cycle_orders)
})

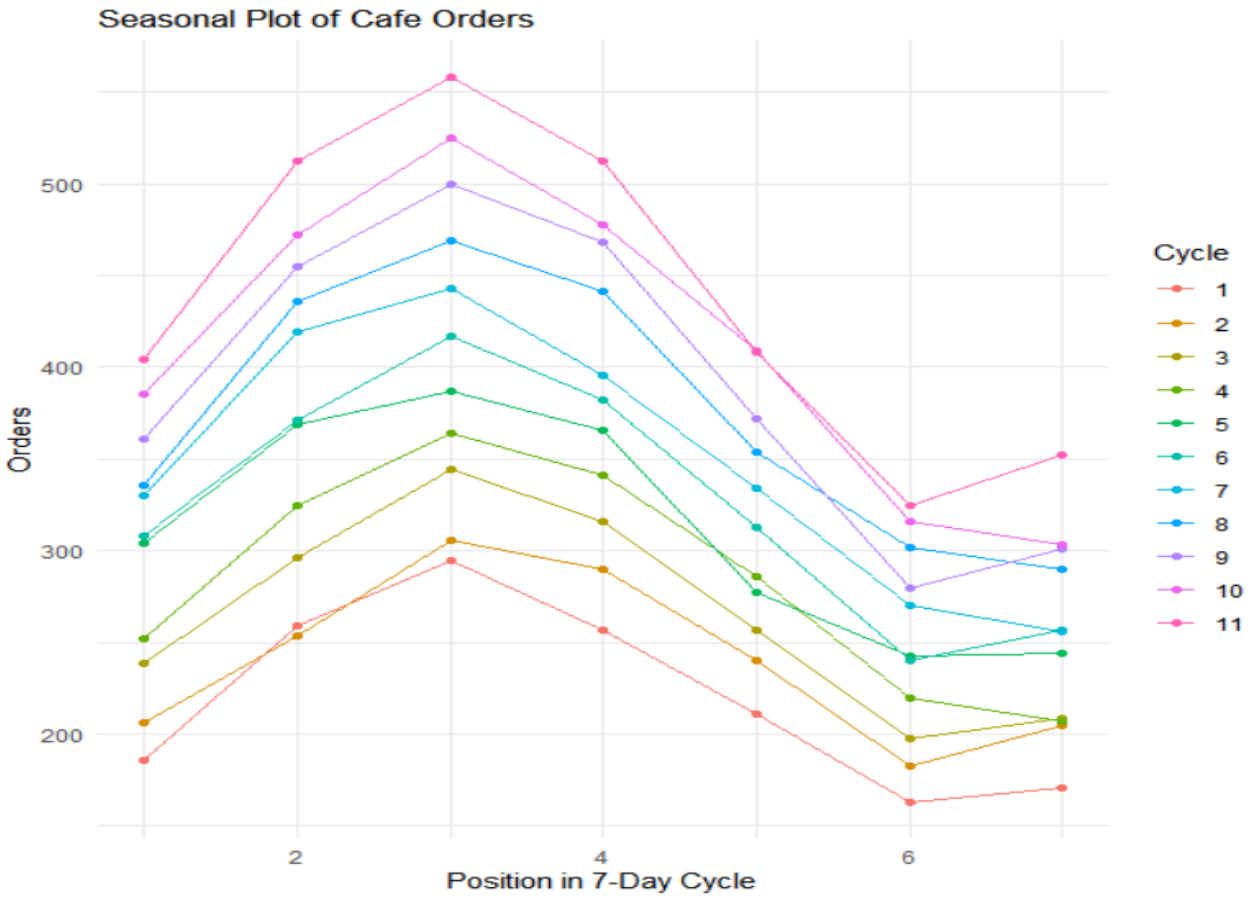
peak_count <- sum(peak_positions == peak_day)
```

```
cat("Seasonal peak position:", peak_day, "\n")

cat("Cycles with highest order at peak position:",

    peak_count, "out of 11\n")
```

Output



Answer and interpretation:

The seasonal plot compares all 11 cycles. If most lines rise and fall at similar positions, the seasonal pattern is consistent. The printed count tells how many cycles have their highest observed order on the seasonal peak position identified in Task 2.

5. Task 4 — Confirm seasonality using autocorrelation

Code:

```
d <- resid(lm(orders ~ t))

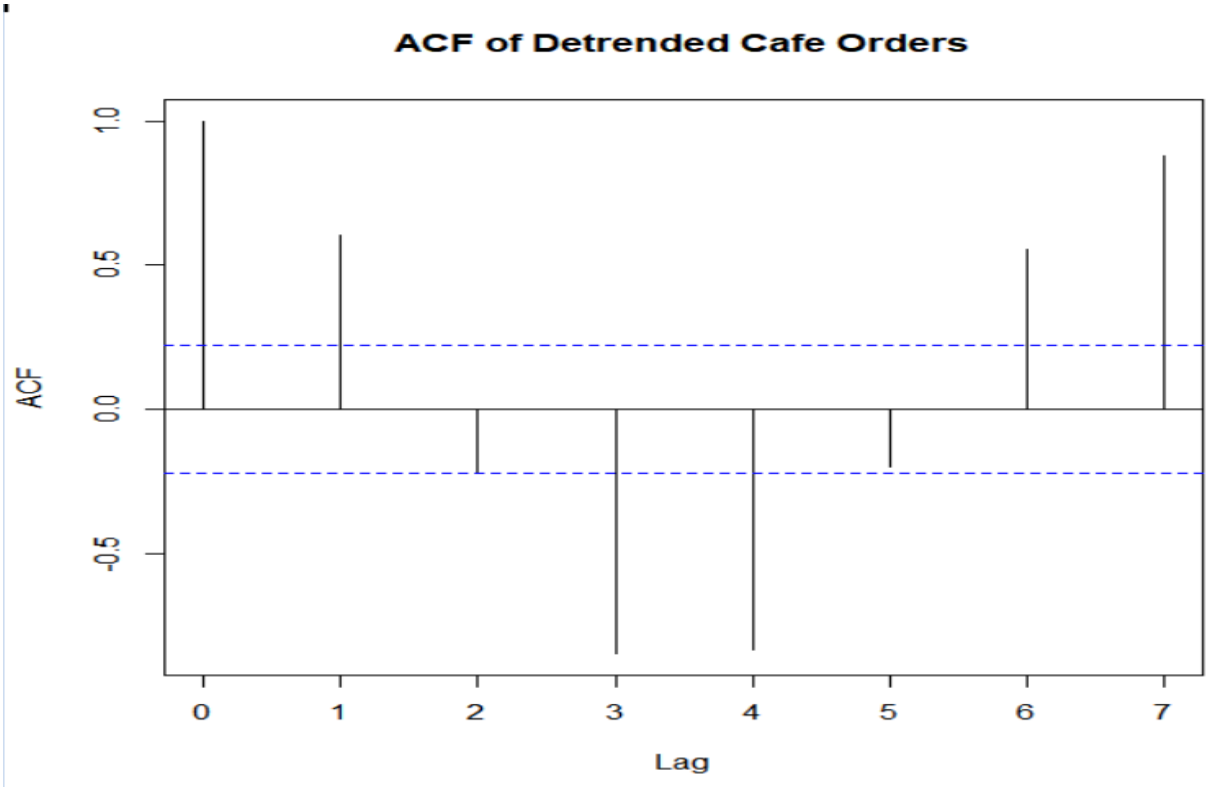

a <- acf(d, lag.max = 7, plot = FALSE)
acf_values <- as.numeric(a$acf)


lag7 <- acf_values[8]
lag3 <- acf_values[4]
bound <- 2 / sqrt(n)


cat("Significance bound:", bound, "\n")
cat("Autocorrelation at lag 7:", lag7, "\n")
cat("Autocorrelation at lag 3:", lag3, "\n")


plot(a, main = "ACF of Detrended Cafe Orders")
```

Output



Important: R includes lag 0 as the first ACF value, so lag 7 is element 8 and lag 3 is element 4.

Interpretation:

- Significance bound: $2/\sqrt{77} \approx 0.228$.
- Lag 7 checks whether observations seven days apart are correlated, which supports a weekly seasonal pattern if the value is meaningfully positive and exceeds the bound in absolute value.
- Lag 3 checks the relationship between observations three days apart. A value outside the significance bounds indicates notable autocorrelation at that lag.

Compare the two printed autocorrelations with $+\ 0.228$ and $-\ 0.228$ to decide which are statistically notable under this approximate rule.

6. Task 5 — Additive or multiplicative seasonality?

A. Compare the first and last cycle ranges (2 marks)

Formula:

$$Range = Maximumorders - Minimumorders$$

$$Ratio = \frac{Lastcyclerange}{Firstcyclerange}$$

Code:

```
first_cycle <- orders[1:7]

last_cycle <- orders[71:77]


range_first <- max(first_cycle) - min(first_cycle)

range_last <- max(last_cycle) - min(last_cycle)

range_ratio <- range_last / range_first


cat("First cycle range:", range_first, "\n")

cat("Last cycle range:", range_last, "\n")

cat("Range ratio:", range_ratio, "\n")
```

```
if (range_ratio > 1.2) {
  cat("Decision: Multiplicative seasonality is likely.\n")
} else {
  cat("Decision: Compare the ranges and plot before deciding.\n")
}
```

Answer: The series is generated using a cosine seasonal factor multiplied by an increasing trend. Therefore, the seasonal swings are expected to grow as the level of orders increases, supporting a multiplicative seasonal model. The computed ranges and ratio should be reported from R.

B. Remove seasonality and calculate the adjusted trend (2 marks)

For multiplicative decomposition:

$$Adjustedorders = \frac{Orders}{Seasonalindex}$$

Code:

```
dc <- decompose(x, type = "multiplicative")

adjusted <- as.numeric(x) / rep(dc$figure, length.out = n)

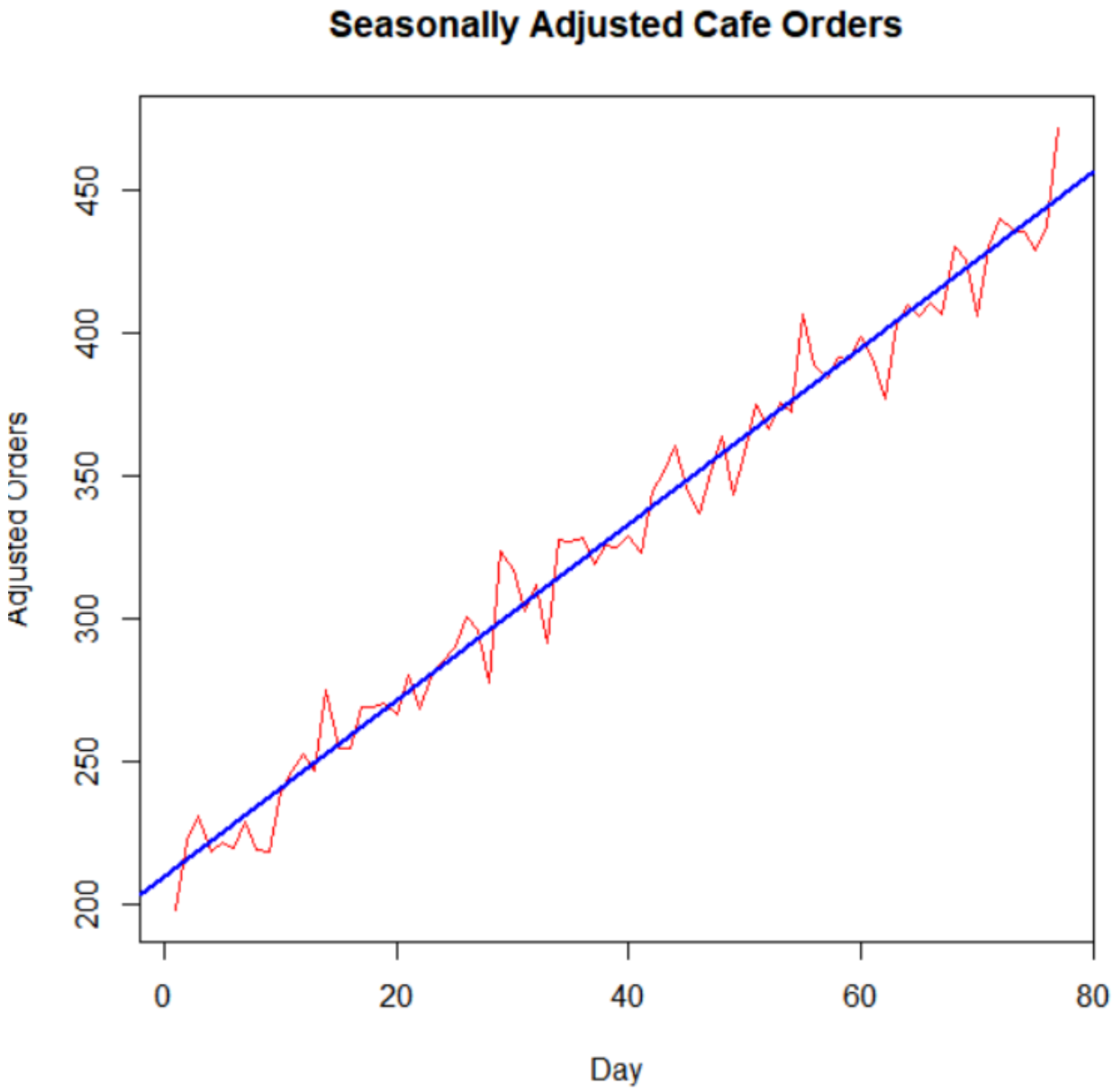
adjusted_fit <- lm(adjusted ~ t)
adjusted_slope <- coef(adjusted_fit)[2]

cat("Adjusted trend slope per day:",
    adjusted_slope, "\n")

plot(t, adjusted, type = "l", col = "red",
     main = "Seasonally Adjusted Cafe Orders",
     xlab = "Day",
     ylab = "Adjusted Orders")

abline(adjusted_fit, col = "blue", lwd = 2)
```

Output



Answer: The adjusted trend slope is the coefficient of `t` printed by R. It represents the estimated daily change in cafe orders after removing the repeating seasonal effect.

C. Advice to management

Advice: Management should plan staffing and ingredient purchases around the recurring seven-day order pattern while preparing for increasing overall demand.

7. Final result

The daily cafe order series contains 77 observations across 11 complete seven-observation cycles. The supplied formula suggests an upward trend and multiplicative seasonality. The R plots and printed statistics should be used to confirm the peak and trough positions, cycle consistency, autocorrelations, range ratio, and seasonally adjusted trend.

Code of Conduct: I certify that this work is my own.

Signature: *Udhayakumar*